

Electroencephalogram

Electroencephalogram

- The term electroencephalogram (EEG) (introduced by the German psychiatrist Hans Berger) refers to record of spontaneous electrical activity of the brain taken from the surface of scalp
- The spontaneous electrical activity of the brain is due largely to graded or summated postsynaptic potentials in the many hundreds or thousands of brain neurons that underlie the recording electrode at the surface of scalp.
- The electrical activity of the brain can also be recorded from the pial surface of the brain cortex after opening the skull (e.g. during brain surgery). The term electrocorticogram (ECoG) is used to denote such a record.

Terms

- Electroencephalography
- Electroencephalogram

Technique of EEG recording

- **Bipolar** versus **unipolar** electroencephalogram
- **Bipolar electroencephalogram** is the record of potential variations between a pair of scalp (cortex) electrodes.
- **Unipolar electroencephalogram** is the record of potential variation between an active scalp (cortical) electrode and a theoretically indifferent electrode placed on some part of the body (distant from the cortex such as earlobe).

- the electroencephalography basically involves record of potential difference between a pair of scalp electrodes (bipolar electroencephalography) or between an active scalp electrode and an indifferent electrode (unipolar electroencephalography).
- The electrodes are connected to a cathode ray oscilloscope (CRO) via a suitable amplifier. The modern EEG machine has 8–16 channels, which can simultaneously record electrical activity from areas of many scalp electrodes.

Location of electrodes

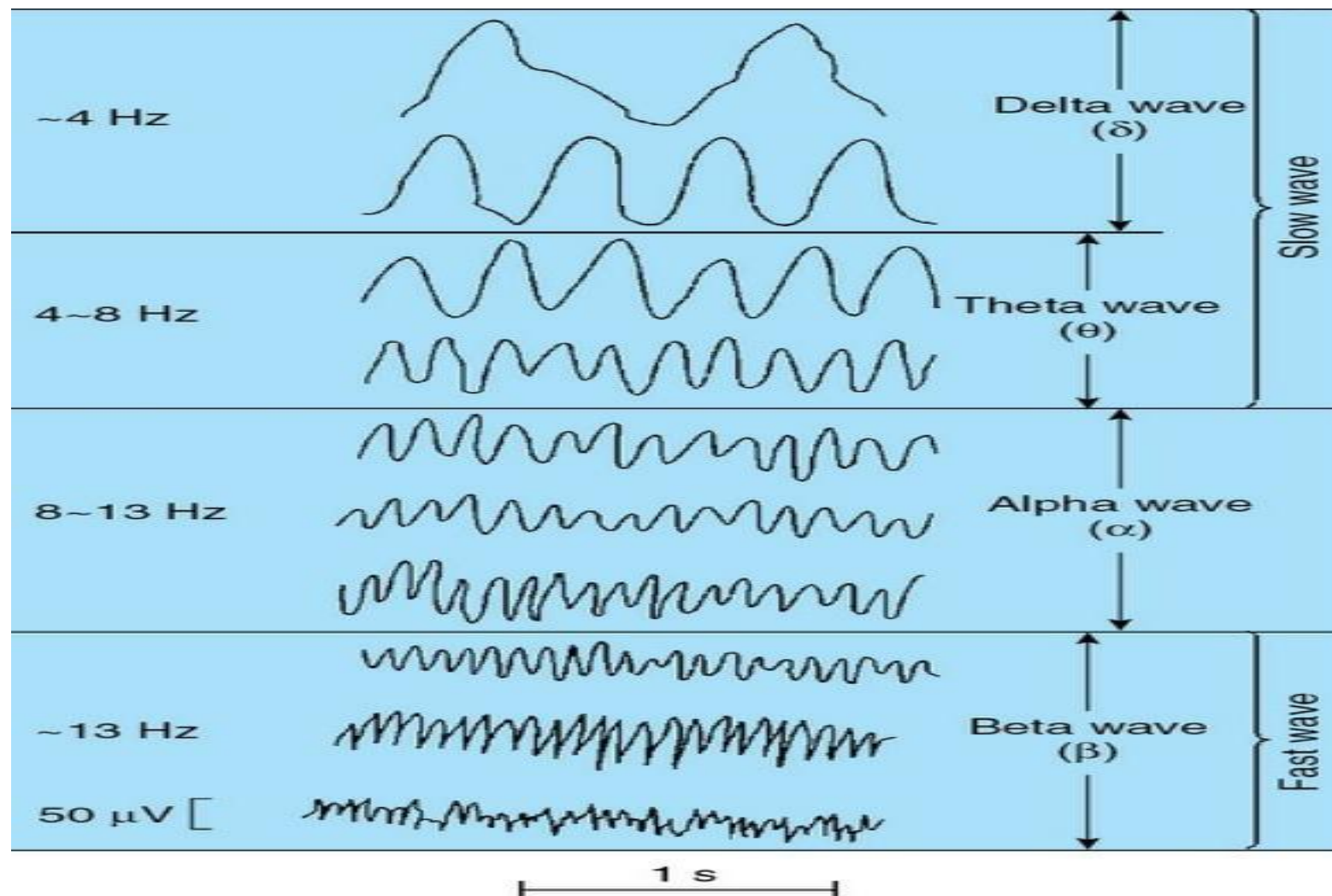
- Different techniques such as Illinois, Montreal, AIRD or Cahn methods have been described to determine the electrode attaching positions
- The most popular method is the International 10–20 system (Montreal method)
- This system is based upon measurement from four standard points on the head: the nasion, the union and the left and right preauricular points. This system is named '10–20 system', because electrodes are spaced either 10 or 20% of the total distance between a given pair of skull landmarks

Normal electroencephalogram

- The electroencephalogram (EEG) consists of waves which are oscillations in the electrical potential of the brain having the following characteristics:
- The oscillations differ in the frequency and amplitude at different points on the scalp and during different stages of mental alertness.
- Frequencies of brain waves range from 1 to over 50 cycles/s
- Amplitude of brain waves may vary from 50 to 200 μV .
- Much of the time, brain waves are irregular and no general pattern is obtained. At other times, distinct patterns do appear.
- Different waves recorded in a normal person, depending on their frequency are classified as alpha, beta, theta and delta waves

Classification of Brain Waves Depending on Frequency of Oscillations

Type of EEG wave	Frequency (in Hz)	Amplitude (in μV)
Delta (δ)	1-4	20-200
Theta (θ)	4-7	10
Alpha (α)	8–13	50
Beta (β)	14–30	5–10



Alpha waves

- These are the most prominent component of EEG obtained from adult humans who are awake but quiet and at rest with the eyes closed. Alpha waves are said to result from spontaneous activity of nonspecific thalamocortical system.

Characteristic features of alpha waves are:

- Frequency of alpha waves varies from 8 to 13 Hz.
- The mean peak frequency in a large number of normal adults is 4.2 Hz.
- Amplitude of these waves slowly waxes and wanes, but the average amplitude is about 50 μ V.
 - Location. Alpha waves are most marked in the parietooccipital area of the scalp, though these are observed sometimes from other locations as well.
 - Disappear during sleep.

Causes of decreased frequency of alpha waves are:

- Old age, due to decreased cerebral perfusion leading to decreased cerebral metabolism,
- Low blood glucose level,
- Low body temperature
- Low levels of adrenal glucocorticoids,
- High arterial partial pressure of CO₂ and
- Sleep.
- Encephalopathies (metabolic, toxic or due to hyponatraemia and vitamin B12 deficiency)
 - Acute intoxication with alcohol, drugs like amphetamine, barbiturates, phenytoin and antipsychotic drugs.

Causes of increased frequency of alpha waves are

- High blood glucose level
- Rise in body temperature
 - Low arterial $p\text{CO}_2$ - Forced overbreathing to decrease $p\text{CO}_2$ is sometimes used clinically to elicit latent EEG abnormalities.
- High levels of adrenal glucocorticoids and
- Alerting states.

- The term aroused or alerting response is also used to denote an alpha block, since it is correlated with arousal or alerting response.
- The term desynchronization has also been suggested for alpha block, because it represents breaking up of the obviously synchronized neural activity necessary to produce regular alpha waves.
- However, the term desynchronization is misleading as the fast EEG activity seen in alert states is also synchronized, but at a higher rate.

Alpha block Alpha block

- Alpha block or alpha attenuation refers to a phenomenon in which the alpha waves attenuate and are replaced by the fast, irregular waves of low amplitude
- Alpha block occurs when:
 - The persons open their eyes
 - When the individual engages in conscious mental activity such as doing mathematical calculations and
 - When any form of stimulation is applied

Beta waves

- Frequency of beta waves is usually between 14–30 cycles/s.
- Amplitude (voltage) of beta waves (5–10 μV) is much lower than alpha waves (50 μV).
- Location. They are frequently recorded from the parietal and frontal regions.

Seen under following conditions:

- Tension and CNS activation,
 - Arousal response (or alpha block),
 - Infants have fast beta-like activity in EEG and occipital rhythm is slow 0.5–2/s patterns and
- #Barbiturates induce beta activity typically at a frequency of 18–24 Hz.

Theta waves

- Theta (θ) waves usually do not occur in normal waking individuals (except in newborn infants).
- Theta component persists in adult life in 10– 15% of normal subjects. Usually theta waves are seen under following conditions: • Emotional stress in adults, particularly during disappointment and frustration. • Many brain disorders. • Theta component of EEG often accentuates during crying in children.

Characteristic features of theta waves are:

- Frequency is between 4–7 Hz.
- Amplitude (10 μ V) is slightly larger than alpha waves.

#Location :They are recorded from the temporal and parietal region in children.

Delta waves

- Delta waves do not occur in normal waking individuals.

These are seen in following conditions:

- Deep sleep (stages III and IV of non-REM sleep),
- Infancy and
- Serious brain damage.

Characteristic features are:

- Frequency is less than 4 Hz

Amplitude is very high (20–200 μ V),

- Can be produced by overbreathing,
 - Occurs strictly in the cortex independent of activities in lower regions of the brain, therefore, they occur in sleep when cortex is released from the activating influence of lower centres.

Variations in the EEG wave formation with age

- The EEG, wave forms at rest in humans and varies with age as
 - In infants (up to 1 year of age) the occipital rhythm is slow (0.5–2 Hz) than those of adults (8–13 Hz).
 - In children, the occipital rhythm speeds up and, the adult alpha pattern gradually appears during adolescence.
 - In children of 4 years of age, theta and alpha waves of 7– 8 Hz appear.
 - In 9-year-old children, an alpha wave of about 10 Hz appears in the occipital region.
 - After 15 years of age, the EEG waveforms become almost the same as those of adults.
 - In adults (between 15 and 60 years) the EEG wave forms and the typical adult alpha pattern is seen. However, even during these ages, the slow waves decrease as the age rises and the fast waves increase over 60 years of age, the slow waves increase again as the age increases.

Applied aspects

- Abnormal EEG waveforms

1. Epilepsy

2. Consciousness dysfunction

A slow wave appears in the case of consciousness dysfunction according to the degree of seriousness.

In the serious case, a general slow wave or delta-wave prevails in the EEG waveform. In the more serious cases, the waveform becomes flat or sometimes disappears.

However, there are many exceptions. Disturbances of consciousness include coma, syncope and stupor.

3. Organic brain dysfunction

4. Brain death Labelling brain death is important to fulfill the desire to obtain organs for transplant operations and to remove the life-support system.

Criteria for labelling brain death are;

when the brain cells stop activity,

the patient falls into a comatose state and finally respiration stops,

the pupils are dilated,

the EEG waveform becomes flat in all channels and finally disappears.

Most patients die in this state.